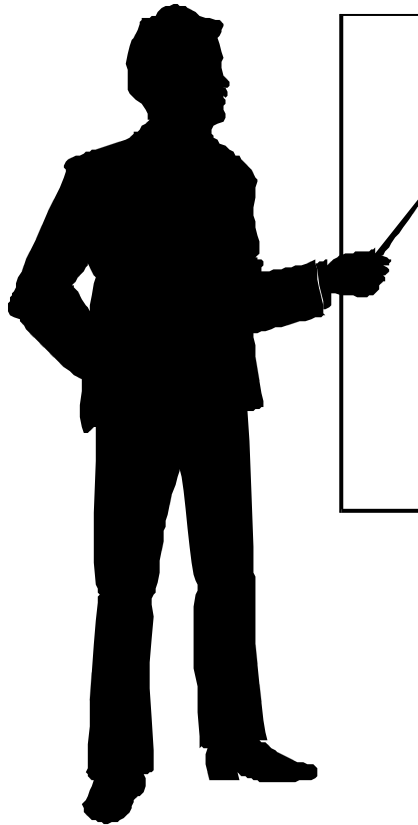




Routing Protocols (Part 1: RIP & Router Configurations)

Introduction

- An internet is a combination of networks connected by routers
- A metric is a cost assigned for passing through a network.
 - ◆ the total metric of a particular route is equal to the sum of the metrics of networks that comprise the route.
 - ◆ the router chooses the route with the shortest (smallest) metric
- RIP (Routing Information Protocol): treating each network equals.
 - ◆ The cost of passing through each network is the same.
 - so if a packet passes through 10 networks to reach the destination, the total cost is hop counts.

Introduction

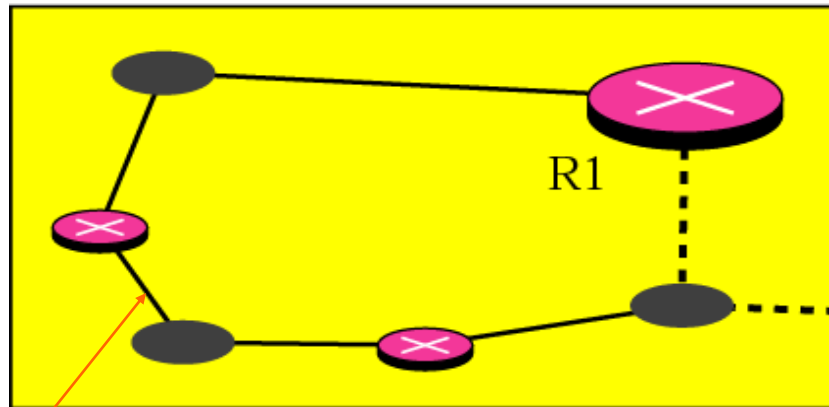
- **OSPF (Open Shortest Path First)**
 - ◆ Allowing the administrator to assign a cost for passing through a network based on the type of service required.
 - ◆ A route through a network can have different costs (metrics)
- **BGP (Border Gateway Protocol)**
 - ◆ Criterion is the policy, which can be set by the administrator.
 - ◆ Policy defines what paths should be chosen.
- **Static and Dynamic tables**
- **Unicast Routing and Multicast Routing**

Intra and Extra Domain Routing

- Because an internet can be so large, one routing protocol cannot handle the task of updating routing tables of all routers.
- So, an internet is divided into autonomous systems.
- An autonomous system (AS) is a group of networks and routers under the authority of a single administration.
- Intradomain routing
 - ◆ used for the routing inside an autonomous system
- Interdomain routing
 - ◆ used for the routing between autonomous systems

Autonomous Systems

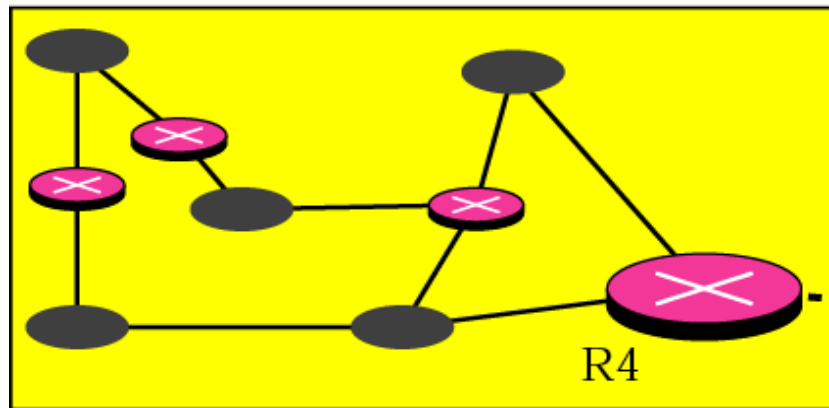
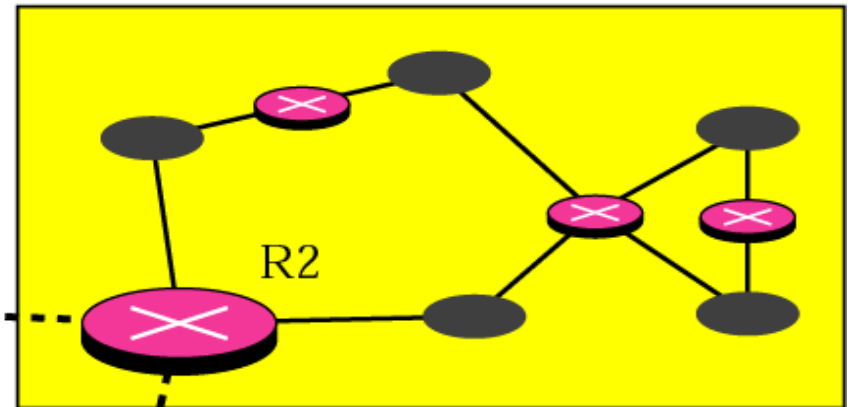
Autonomous system



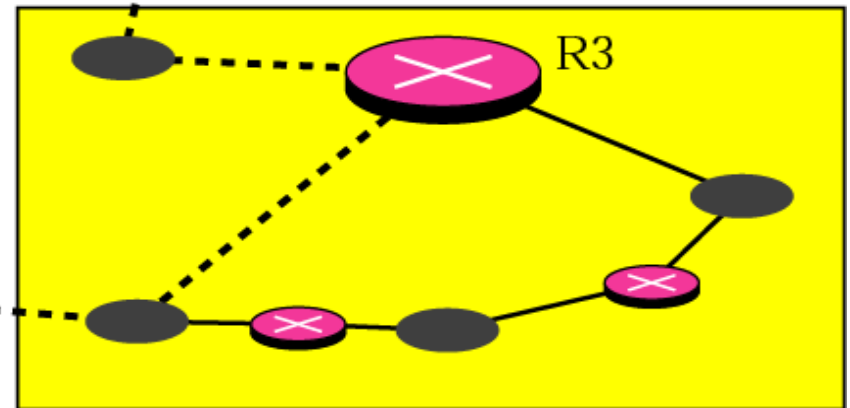
Intradomain routing

Interdomain routing

Autonomous system



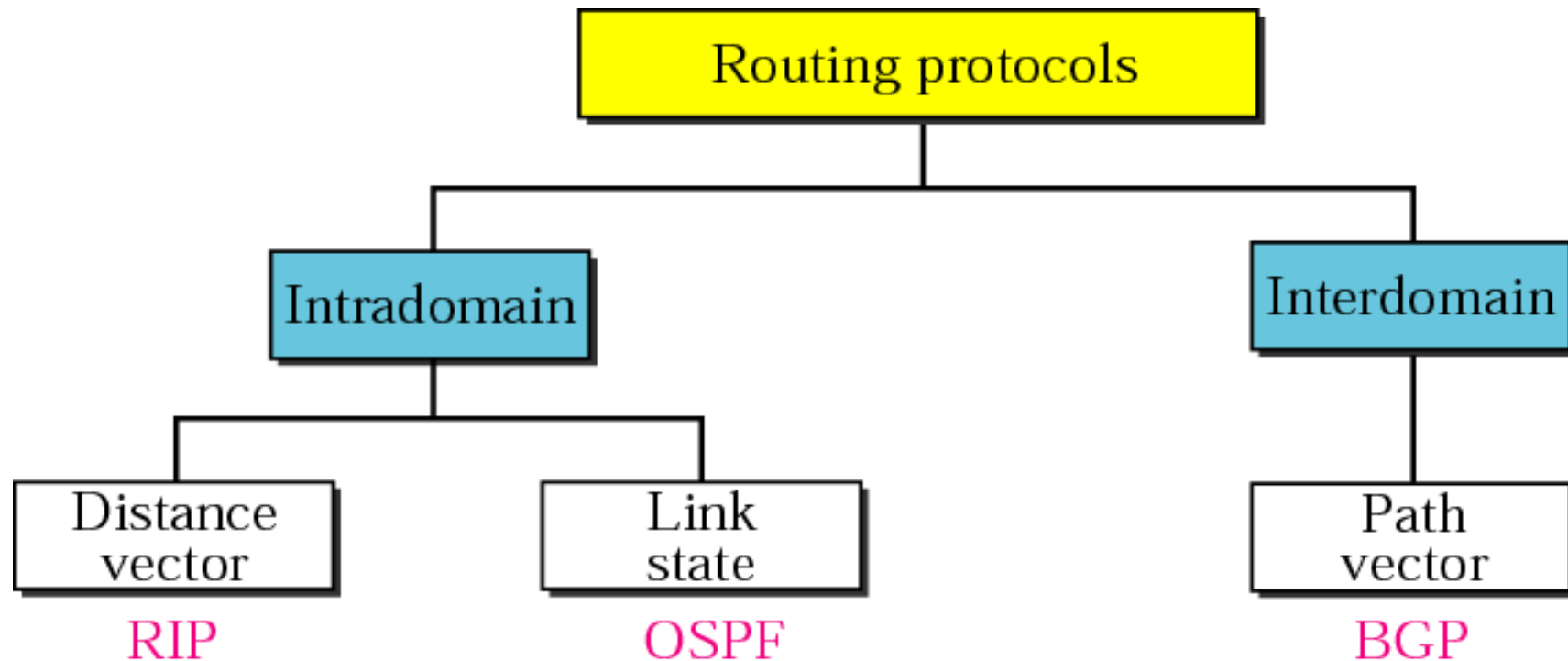
Autonomous system



Autonomous system

Intra and Extra Domain Routing (Cont'd)

Popular routing protocols



Distance Vector Routing

- In distance vector routing, the least cost route between any two nodes is the route with minimum distance. In this protocol each node maintains a vector (table) of minimum distances to every node
- Distance Vector Routing
 - ◆ each router periodically shares its knowledge about the entire internet with neighbors
 - ◆ the operational principles of this algorithm
 1. Sharing knowledge about the entire autonomous system
 2. Sharing only with neighbors
 3. Sharing at regular intervals (ex, every 30 seconds)

Distance Vector Routing Tables

To Cost Next

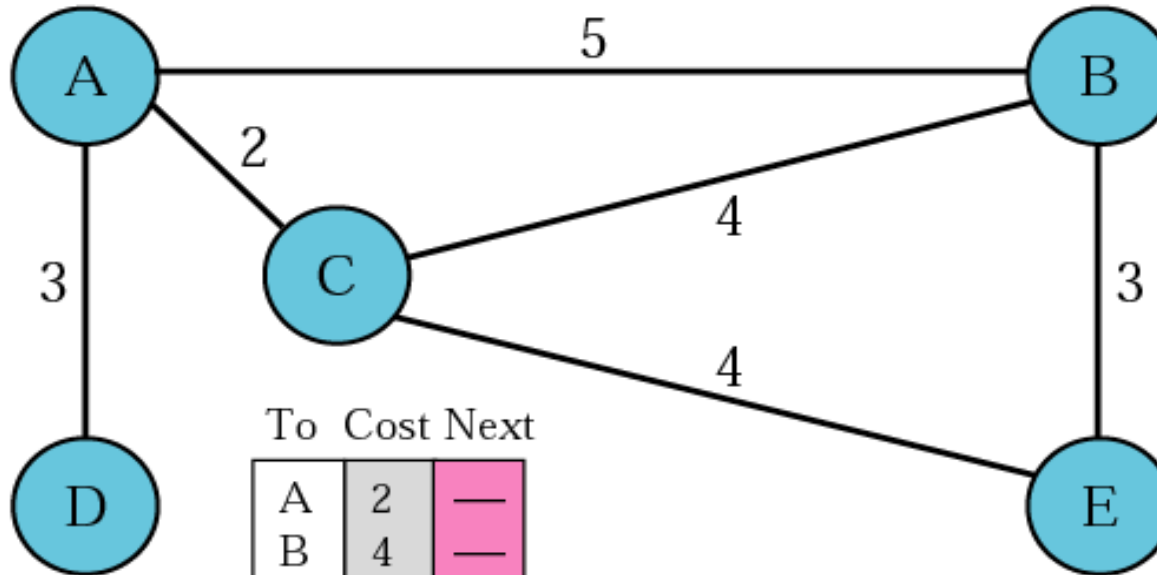
A	0	—
B	5	—
C	2	—
D	3	—
E	6	C

A's table

To Cost Next

A	3	—
B	8	A
C	5	A
D	0	—
E	9	A

D's table



To Cost Next

A	2	—
B	4	—
C	0	—
D	5	A
E	4	—

C's table

To Cost Next

A	5	—
B	0	—
C	4	—
D	8	A
E	3	—

B's table

To Cost Next

A	6	C
B	3	—
C	4	—
D	9	C
E	0	—

E's table

Distance Vector Routing

- In the previous slide, we have seen Distance Vector Routing

Tables A, B, C, D, E

- In the next two slides, we will see how the Table A was formed.
- Similarly, other tables are formed.

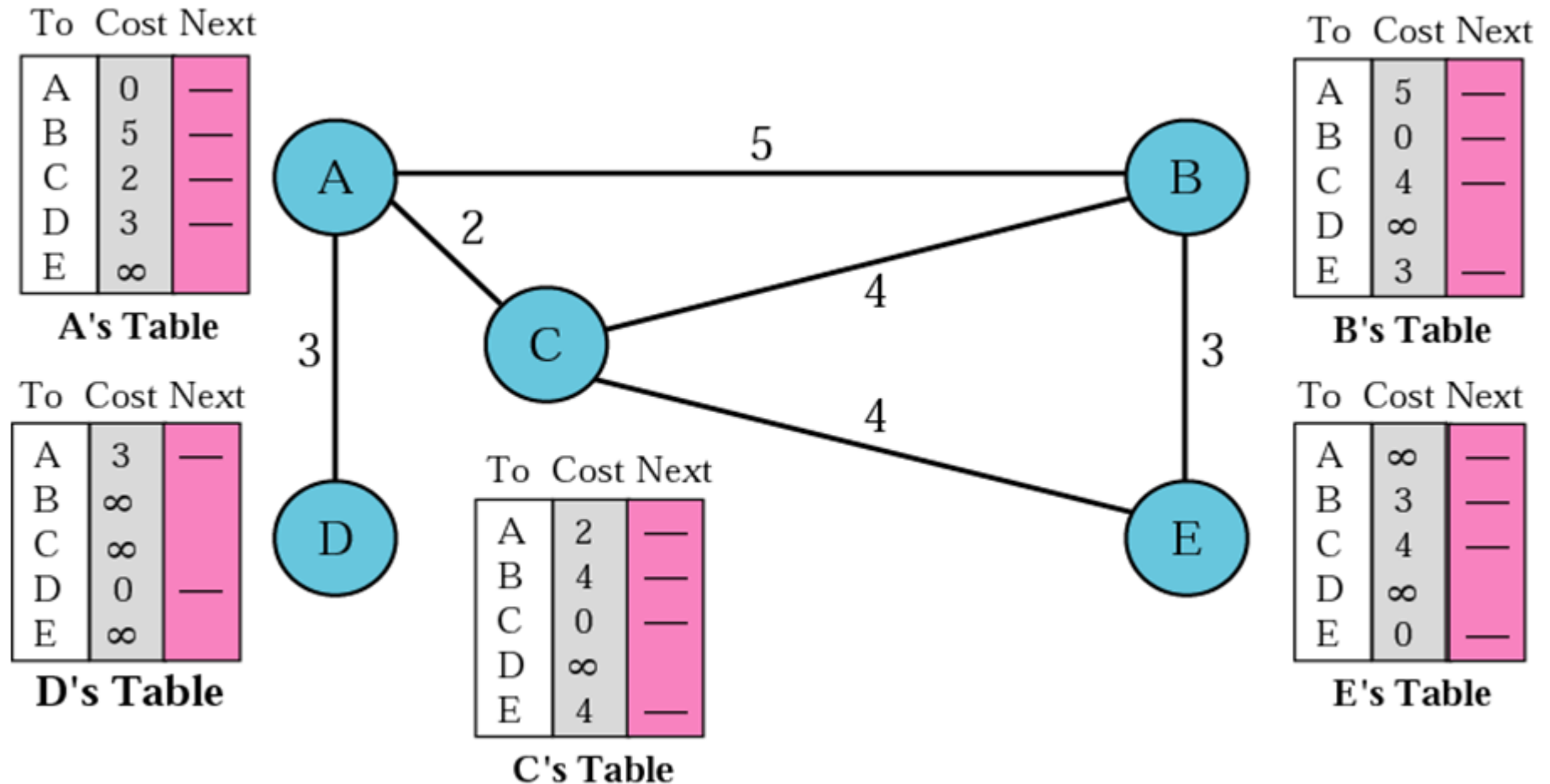
Distance Vector Routing

Step-by-Step Formation:

- **Initial State:** Each router begins with a simple table of directly connected neighbors.
- **Information Exchange:** Every router sends its routing table to its immediate neighbors.
- **Updating Tables:** Each router updates its table based on new information received.
- **Repeating Updates:** Routers continue exchanging and updating data until all have the shortest paths to every other router.

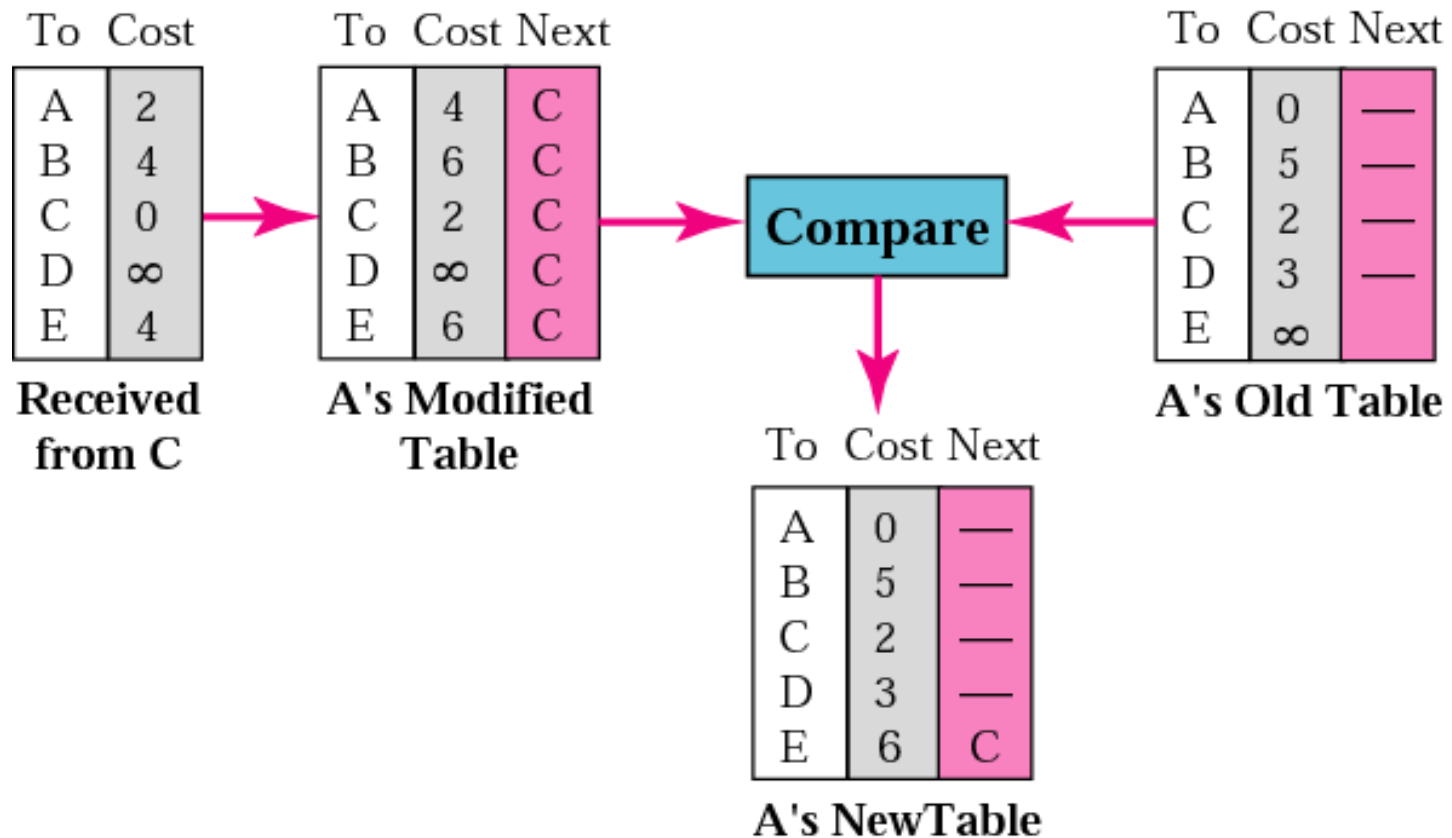
This process repeats periodically, ensuring routers have updated paths

Initialization of Tables in Distance Vector Routing



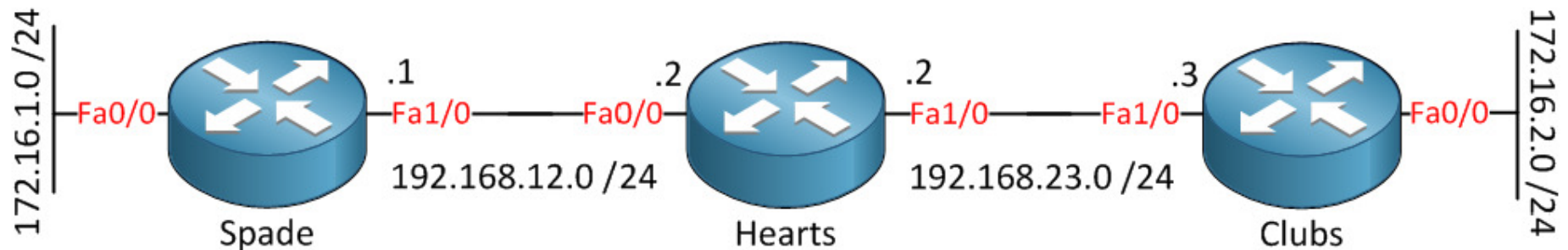
Updating in Distance Vector Routing

- In distance vector routing, each node shares its routing table with its immediate neighbors periodically and when there is a change.



Router configuration (example of CISCO Routers)

RIP (Routing Information Protocol) is one of the routing protocols you need to understand if you want to pass the Cisco CCNA exam. If you have no idea how RIP works I suggest to [read this article](#) first where I explain how RIP works. In this article I'll show you how to configure RIP on a Cisco router. This is the topology that I will use:



Above we see 3 routers called Spade, Hearts and Clubs. There are a couple of networks so we'll have something to advertise in RIP. First let's configure all the interfaces:

Router configuration cont...

```
Spade>enable
```

```
Spade#configure terminal
```

```
Spade(config)#interface fastEthernet 0/0
```

```
Spade(config-if)#no shutdown
```

```
Spade(config-if)#ip address 172.16.1.1 255.255.255.0
```

```
Spade(config-if)#exit
```

```
Spade(config)#interface fastEthernet 1/0
```

```
Spade(config-if)#ip address 192.168.12.1 255.255.255.0
```

```
Spade(config-if)#no shutdown
```

Router configuration cont...

```
Hearts>enable
Hearts#configure terminal
Hearts(config)#interface fastEthernet 0/0
Hearts(config-if)#no shutdown
Hearts(config-if)#ip address 192.168.12.2 255.255.255.0
Hearts(config-if)#exit
Hearts(config)#interface FastEthernet 1/0
Hearts(config-if)#no shutdown
Hearts(config-if)#ip address 192.168.23.2 255.255.255.0
Hearts(config-if)#exit
```

Router configuration cont...

```
Clubs>enable
Clubs#configure terminal
Clubs(config)#interface fastEthernet 0/0
Clubs(config-if)#no shutdown
Clubs(config-if)#ip address 172.16.2.3 255.255.255.0
Clubs(config-if)#exit
Clubs(config)#interface fastEthernet 1/0
Clubs(config-if)#no shutdown
Clubs(config-if)#ip address 192.168.23.3 255.255.255.0
Clubs(config-if)#exit
```

Router configuration cont...

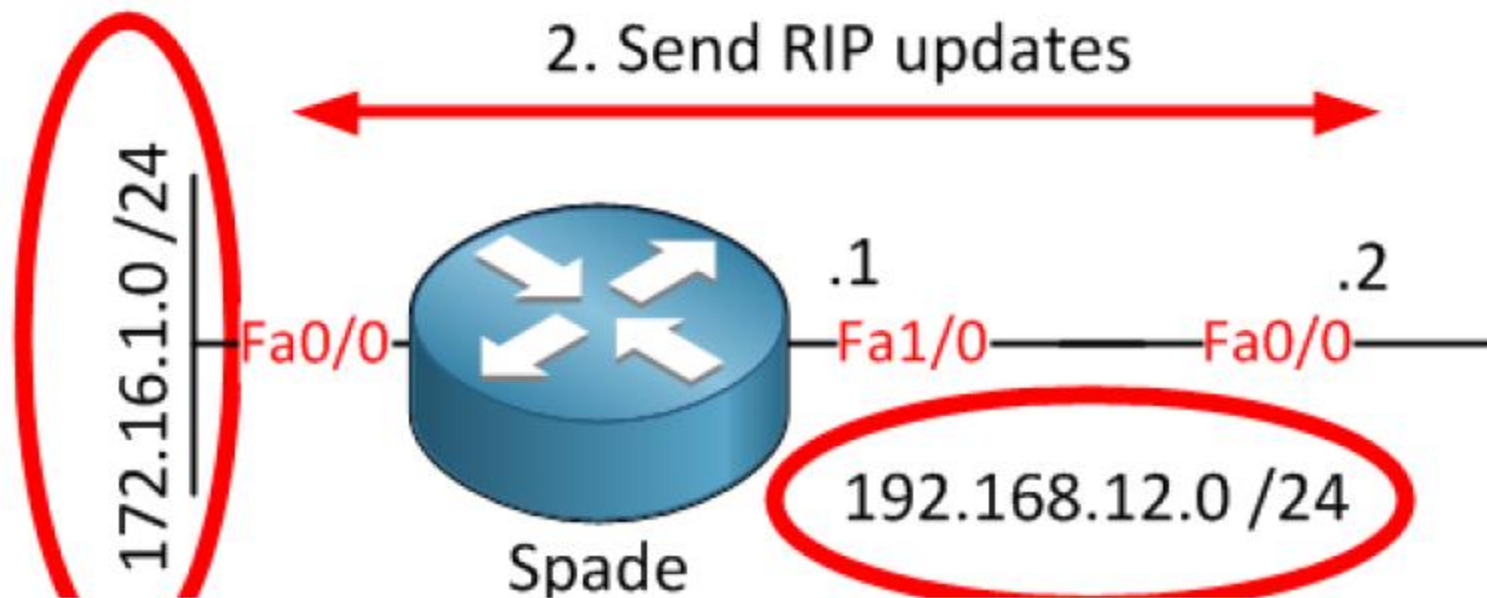
```
Spade(config)#router rip
Spade(config-router)#network 192.168.12.0
Spade(config-router)#network 172.16.1.0
```

```
Hearts(config)#router rip
Hearts(config-router)#network 192.168.12.0
```

We use the **router rip** command to go to the RIP configuration. Next step is to use the **network** command which does two things:

Router configuration cont...

How to configure RIP on a Cisco router | Networklessons.com



Some important commands of CISCO & Huawei routers

Cisco Command	Huawei Equivalent	Description
<code>enable</code>	<code>system-view</code>	Enter privileged mode or system view.
<code>configure terminal</code>	<code>system-view</code>	Enter global configuration mode.
<code>interface fastethernet 0/0</code>	<code>interface GigabitEthernet 0/0/0</code>	Configure an interface (Huawei typically uses GigabitEthernet instead of FastEthernet).
<code>no shutdown</code>	<code>undo shutdown</code>	Enable the interface.
<code>exit</code>	<code>quit</code>	Exit the current mode.
<code>router rip</code>	<code>rip</code>	Enter RIP routing configuration mode.
<code>network <network></code>	<code>network <network></code>	Specify the networks to include in RIP.
<code>show ip route</code>	<code>display ip routing-table</code>	Display the routing table.
<code>router ospf 1</code>	<code>ospf 1</code>	Enable OSPF routing protocol.